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TPPProtecting.2.1

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Protecting a Test Program

Protecting a Test Program

The Test Program Protection Tool performs the following actions:

- | | |
|----|---|
| 1. | Selects files and folders to protect and copy. |
| 2. | Assigns a destination directory. |
| 3. | Encrypts protectable files and copies unprotectable files, placing everything in the destination directory. |
| 4. | Writes a log of its actions. |

The following sections describe the steps for using the Test Program Protection Tool to create a protected version of an existing test program:

- | | |
|---|---|
| • | Before You Begin |
| • | Starting the Test Program Protection Tool |
| • | Using the Display to Protect a Program |
| • | Controls and Labels |
| • | Delivering a Protected Test Program |

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Protecting a Test Program : Before You Begin

Before You Begin

Complete test program development before you protect a test program. While you can reapply protection to your source test program, you cannot edit a protected test program. Therefore, you should protect a test program only after you finish making changes to it.

Export your test program’s Microsoft Excel workbook to ASCII files. (For more information, see [Preparing DataTool Worksheets](#).) Compile all digital patterns before including them in a protected test program. Decide which files are required parts of your test program; you can then use the Test Program Protection tool to select these files, or copy the essential files to a “staging” directory and select all files in that directory.

Make sure that you have sufficient space on your destination drive to hold all copied test program elements, and that you have write privileges to the destination folder.

For more information, see [Working With Test Program Files](#).

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Protecting a Test Program : Starting the Test Program Protection Tool

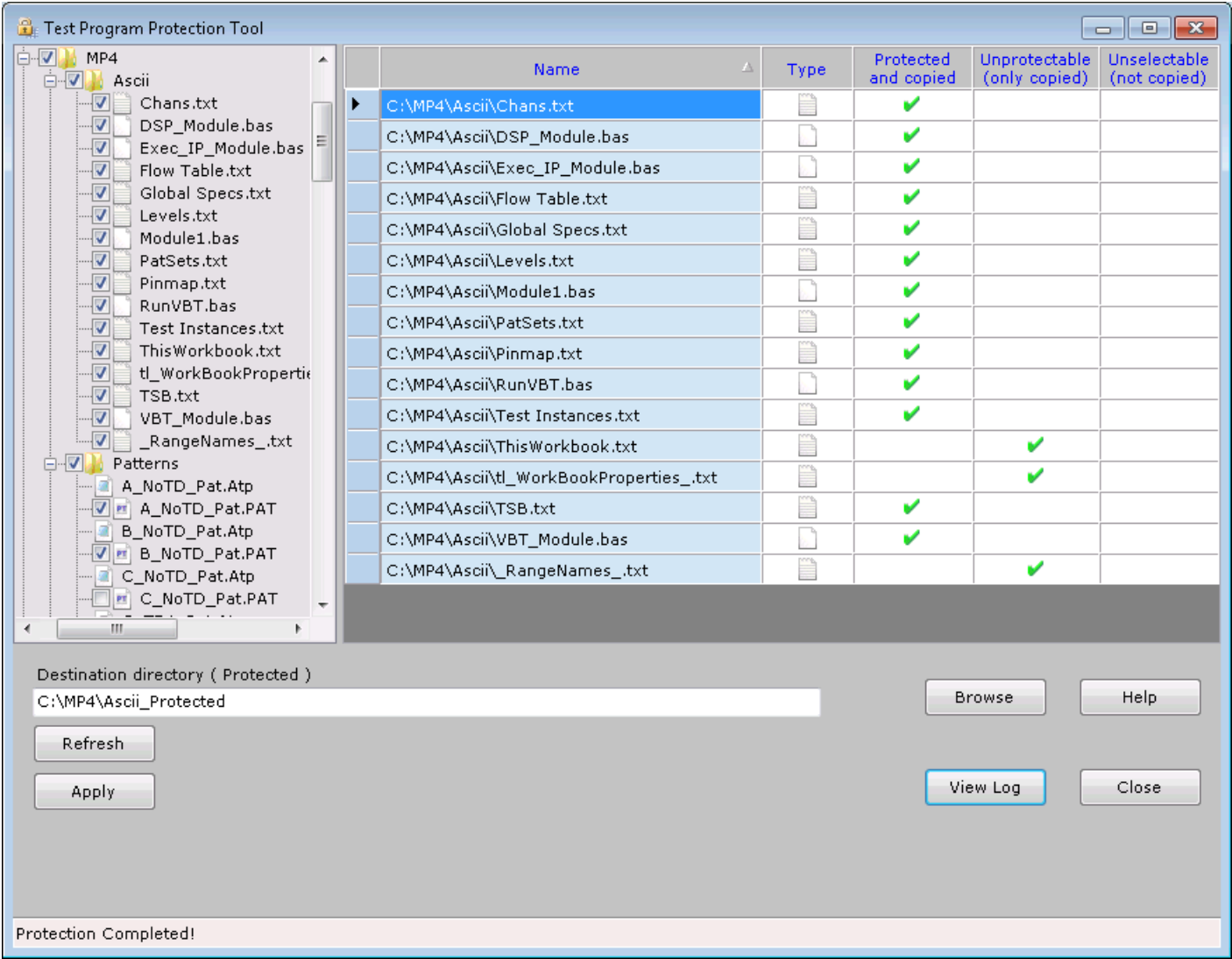
Starting the Test Program Protection Tool

To start the tool from DataTool, on the File tab and IG-XL subtab, click the Protect Test Program button.

You can also start the tool from the Windows Start menu:

Start>All Programs>Teradyne IG-XL V[version]>>Tools>Test Program Protection Tool, where [version] is the version of IG-XL you are running.

The following figure shows the Test Program Protection Tool display.



Test Program Protection Tool

The Test Program Protection Tool works best with a screen resolution of at least 1280x1024 for desktop computers or of at least 1440x900 for laptop computers. Some parts of the tool may be obscured if you use a lower screen resolution, although you can resize the window so that you can use it at lower resolutions.

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Protecting a Test Program : Using the Display to Protect a Program

Using the Display to Protect a Program

To protect an existing test program in ASCII format:

1.

Use the file selection pane to select the folder that contains the test program you want to protect.

If you created a “staging” folder, select that folder instead.

The Test Program Protection Tool reads the contents of the folder and automatically selects (checks) all selectable files in this directory and any subdirectories.

Deselect any files that are not required to run your test program. You can remove (uncheck) files from the selection, but you cannot add files. Deselected files are not protected or copied to the destination. You can press Space to toggle the selection of a highlighted file.

For a description of selectable and unselectable files, see [Recognized File Types](#).

The Test Program Protection Tool’s file selection process does not support:

- Network browsing, except through mapped drives or Network Places; to browse a network resource, assign a drive letter to the resource or use the Add Network Place item to create a Network Place
- Shortcuts; to include a file in your test program, copy it into the source folder; files not located in the test program folder or its subfolders will not be protected
2.

Enter the path and name of the destination folder. The tool stores the protected and copied files in this folder.

The tool suggests the name of the source folder with _Protected appended. You can accept the suggestion, type a new destination folder, or click Browse to select another.

If the folder you specify does not exist, the tool asks permission to create it. If you click Yes, the protection process continues; if you click No, the protection process stops.

If you specify an existing folder, it must be empty. The tool does not overwrite existing files.

The destination folder must be outside the source folder.

You can type UNC-style pathnames, which have the form \\server\resource\path.

3.

Click Apply to begin protecting and copying files.

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Protecting a Test Program : Controls and Labels

Controls and Labels

The Test Program Protection Tool provides the following controls:

Refresh	Rereads the list of files and folders in the file selection pane; maintains existing selections.
Destination directory (protected)	Specifies the folder in which you want the Tool to write the protected test program files.
Browse	Opens a Browse for Folder dialog to help you select a destination folder.
Apply	Begins protecting the selected files and folders.
Abort	Aborts a running protection process. Removes all files from the destination folder but does not affect the source files. Available only during a protection process.
Close	Exits the Test Program Protection Tool.
View log file	Displays the manifest file so you can view a report of the tool's actions. Available only after the protection process completes.

The file selection pane shows the following columns; each selected file has an entry in one column:

Protected and copied	Files with an arrow (➡) in this column can have protection applied and can be stored in the destination folder. After each file is protected and stored, the arrow changes to a checkmark (✅).
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Unprotectable (only copied)	Files with an arrow (↔) in this column can be copied to the destination folder but can not have protection applied. After each file is copied, the arrow changes to a checkmark (✓).
Unselectable (not copied)	Files with a do not enter sign (⊘) in this column cannot be protected and cannot be copied to the destination folder.

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Protecting a Test Program : Delivering a Protected Test Program

Delivering a Protected Test Program

You deliver a protected test program exactly as you would deliver an unprotected test program. Gather all relevant files in the destination folder and distribute them normally (for example, by straight copy or within a compressed archive such as a .tar or .zip file).

☒ **Note:**

You must deliver the manifest file with the protected test program. IG-XL does not open a protected test program if it cannot find the manifest file. For more information, see Manifest File. Test your protected program on a test system before you distribute it. Performing this check can uncover any problems with the protection process or any omissions that you may have made in selecting files to protect or copy.

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Using a Protected Test Program

Using a Protected Test Program

This section describes how to use a test program that has been protected. The following topics are available:

- [Running a Protected Test Program](#)
- [Working With Test Program Files](#)

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Using a Protected Test Program : Running a Protected Test Program

Running a Protected Test Program

You run a protected test program exactly as you run an unprotected test program. IG-XL unscrambles the protected files transparently, and the test program behaves as though unprotected. You can open a protected test program only on a test system using an operator interface.

The following topics are available:

- | | |
|---|----------------------------------|
| • | Opening a Protected Test Program |
| • | Checking for Consistency Errors |

For more information on operator interfaces, see [About Production Controls](#).

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Using a Protected Test Program : Running a Protected Test Program : Opening a Protected Test Program

Opening a Protected Test Program

Because protected test programs include exported ASCII worksheets instead of Microsoft Excel workbooks (.xls) files, you open a protected test program the same way you would open a test program exported to ASCII format.

To open a protected test program from the Operator Interface:

1. Click Test Program Workbook. (Some interfaces label this button Load.)
2. Browse to the directory containing the protected test program.
3. Select Ascii IG-XL Test Program from the Files of type list.
4. Select all files in the file browser pane.

You can do this by clicking one file name, then typing Ctrl+A. You do not need to include the Manifest file, but you can do so if you wish.

5. Click Open to open the test program.

You can customize an operator interface so that it loads a particular test program without requiring intervention by the test floor operator. To do this, specify the test program to load using the TesterControl.LoadThisTestProgram method. To use this method with a protected test program, list all exported, protected files (that is, all files ending in .txt, .bas, .cls, and .frm) in the ProgFiles string. For more information, see LoadThisTestProgram Method.

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Using a Protected Test Program : Running a Protected Test Program : Checking for Consistency Errors

Checking for Consistency Errors

When you try to run a protected test program, IG-XL checks for the following consistency problems:

IG-XL Protected Test Program Consistency Checks

Change	When IG-XL Checks
Protected file (including manifest) modified	For DataTool sheets and VBT modules, during test program load; for pattern files, during validation or first test program run
Protected file (including manifest) missing	During test program load

If the consistency check fails, IG-XL reports the failure and stops loading the test program. You must correct the cause of the failure before you can load the test program.

Changing or removing an unprotected file does not return an error.

Trying to enter Engineering mode with a protected test program loaded reports that the test program is protected and that Engineering mode is not available. To prevent this, you can query a property and disable the button that enables Engineering mode if the property reports that the loaded test program is protected.

To check the protection status in the operator interface, use the `WorkbooksProtected` Property within the `TesterControl` interface.

To check the protection status from a test program, use the read-only `TheExec.WorkbooksProtected` VBT property.

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Using a Protected Test Program : Working With Test Program Files

Working With Test Program Files

Test program development directories frequently accumulate unnecessary files, such as old versions of the test program and patterns. Before you apply protection to a completed test program, determine which files you want to distribute.

The Test Program Protection Tool requires that all of a test program's files, including worksheets and supporting files such as patterns and waveforms, exist in the same folder or its subfolders. If your test program refers to supporting files in a separate folder, copy those files into a subfolder. (Remember to update your test program to refer to the new paths.)

The Test Program Protection Tool does not protect test programs in Microsoft Excel (.xls) format. Instead, it protects DataTool workbooks in ASCII format. Before you protect a test program, you must export its Microsoft Excel workbook to ASCII files. For more information, see [Preparing DataTool Worksheets](#).

You can simplify the protection process by copying all the files your test program requires (such as ASCII DataTool sheets, patterns, and waveforms) to a separate "staging" folder that you create for this purpose. The staging folder should contain only those files required for the test program. Omit any files that you do not want to appear on the tester computer. All files required for the test program must be in this folder or its subfolders. You can then select this folder in the Test Program Protection Tool.

When you select a folder, the Test Program Protection Tool reads the folder's contents and checks to see whether the folder contains a complete, unprotected test program. This check does not guarantee that the test program validates or runs properly. If you later deselect any required files, the tool proceeds with protection of the file and does not warn you, but the protected version of the test program is incomplete and does not work.

If the Test Program Protection Tool finds all elements required for a complete test program, it selects all appropriate files and folders for you. (See [Recognized File Types](#) for a description of how

the tool determines which files and folders to select.)

Test program protection applies to the entire test program and all of its supporting files. You cannot limit protection to parts of a test program. If you modify one part of a test program, such as a digital pattern, and you want to protect it, you must protect the entire test program.

Protecting a test program does not alter any files in the source folder.

Additional topics are covered in the following sections:

- [Preparing DataTool Worksheets](#)

- [Recognized File Types](#)

- [Manifest File](#)

TPPUsing.3.6



Using a Protected Test Program : Working With Test Program Files : Preparing DataTool Worksheets

Preparing DataTool Worksheets

The Test Program Protection Tool operates on DataTool sheets in ASCII format. It does not protect test programs in Microsoft Excel (.xls) format. Before you apply protection to your test program, export all DataTool sheets in your test program's workbook to ASCII files. To export the entire workbook:

1. [Open your test program.](#)

The test program should work properly and should pass IG-XL validation.

2. [Select](#) File>Export Workbook from Microsoft Excel.

3. [Use the](#) Browse for Folder dialog to locate the folder in which you want Excel to save the exported sheets, then click OK.

Excel saves each sheet in a separate ASCII text file, with the extension .txt.

This procedure exports all sheets and VBT modules in the workbook. If your test program contains additional sheets and you do not want those sheets to appear in the version of the test program you distribute, you can export only sheets you wish to distribute. Make sure that you select all sheets and VBT modules that your test program requires. If the protected test program refers to a missing sheet or VBT module, the test program fails.

To export individual worksheets:

1. [Open your test program.](#)

2. [Select a sheet to export.](#)
3. [Select](#) File>Export Worksheet from Microsoft Excel.
4. [Use the](#) Browse for Folder dialog to locate the folder in which you want Excel to save the exported sheet, then click OK.

[Excel saves the sheet in an ASCII text file, with the extension .txt.](#)

5. [Repeat step 2](#) for each sheet you want to distribute.

[For more information, see Importing and Exporting Workbooks and Worksheets.](#)

		
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Using a Protected Test Program : Working With Test Program Files : Recognized File Types

Recognized File Types

The Test Program Protection Tool automatically detects and selects following file types:

- Exported ASCII test program sheets, which must have the extension .txt; the tool distinguishes ASCII test program sheets from normal text files.
- Visual Basic for Test modules, which must have the extension .bas, .cls, or .frm
- Pattern files, which must have the extension .pat or .pat.gz

The tool disables selection of the following file types:

- ASCII pattern files, which are files with the extensions .atp or .atp.gz
- Microsoft Excel workbooks, which are files with the extension .xls

The tool recognizes the following file types and copies them to the destination directory without protection; you cannot change this behavior:

- Nonsheet text files, which are files with the extension .txt but which do not contain DataTool information; this includes configuration files (such as SimulatedConfig.Txt and TesterConfig.Txt)
- Add-ins to Microsoft Excel, which are files with the extension .xla

- Binary Visual Basic for Test files, which are files with the extension .frx
- Wave files, which are files with the extension .wav
- Folders (files within a folder may be protected, but the folders themselves are not)

If you deselect a folder, the tool does not protect or copy any files in that folder or its subfolders. The tool also copies all files of unrecognized type to the destination directory without protection. When the tool copies an unprotected file, it also copies the file’s attributes. For example, a read-only file in the source is also read-only in the destination.

During the protection process, the Test Program Protection Tool copies all relevant (protected and unprotected) files to the destination directory. The tool maintains the subdirectory structure under the source directory, so any patterns and other supporting files appear in the same place in the destination as in the source. The following figure shows how the Test Program Protection Tool works with various types of files.



Test Program Files Before and After Protection

The protection process does not change the names of external files, such as patterns and wave definitions, so references to these files from the original test program work normally. Any data or log files that your program creates during a run are unprotected and function as in an unprotected test program.

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Using a Protected Test Program : Working With Test Program Files : Manifest File

Manifest File

The Test Program Protection Tool creates a log of its activities each time you protect a test program. This log is a text file with a file name in the following format:

Manifest_<Monthname><Date>_<hour> <minute> <second>.txt

where the date and time are the time at which you applied protection to the test program.

This file serves two purposes and appears in two places for each test program you protect:

- In the folder that contains the unprotected source files, the manifest serves as a log of what the tool did to ensure test program protection.
- In the folder containing the protected test program, the manifest serves as a list of what IG-XL must check to enforce consistency protection. When you distribute the protected test program, you must include this file. If you try to run a protected test program without this file, IG-XL reports a consistency check failure and does not open the test program.

 Note:

The manifest must appear with the protected test program. Do not edit or change the manifest after the Test Program Protection Tool writes it. If IG-XL cannot find the manifest or detects that it has been changed, IG-XL refuses to open the test program.

For each file in the protected test program, the manifest contains the following information:

- File name, including directories and relative paths
- Date and time modified

•	File size
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•	Protection applied
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You can use this information to communicate with a production operator about the version of the test program deployed on the test floor and to help determine the cause of any consistency or content errors.

	
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Test Program Protection

Test Program Protection

As a developer, you need to be able to view and change any part of a test program, including the VBT code, patterns, wave files, and DataTool sheets. Once you release a completed and verified test program to production, however, you want to ensure that it runs unmodified on the test floor. Because test programs (Microsoft Excel workbooks, digital patterns, and VBT code) often contain sensitive information, you may want to prevent test floor operators from viewing or changing some aspects of a test program, to protect your company’s intellectual property and to make sure that the test program runs exactly as you verified it, especially if testing occurs at a subcontractor. IG-XL allows you to protect a completed test program to prevent tampering and unauthorized viewing. The Test Program Protection Tool prevents unauthorized access to test programs without requiring passwords, which can be difficult to manage. For example, passwords can travel with former employees, and current employees can forget the password. The Test Program Protection Tool is used with IG-XL Production Mode. Test program protection requires ASCII workbooks, so you must export your test program before applying protection. For more information, see [Preparing DataTool Worksheets](#). Test program protection consists of:

- Content protection

This prevents unauthorized users from opening and viewing vital test program components. Protected test programs and patterns appear scrambled when one views them. Unscrambling the test program components takes place within IG-XL, so a third-party viewer (such as Windows Notepad) or an old version of IG-XL cannot view a protected test program.

- Consistency protection

This ensures that your test program runs exactly as you released it, with no changes or modifications, and that nobody has tampered with any of the related files. For example, users cannot change the function of a test program by substituting their patterns for yours, even if the substitute pattern has the same file name and date stamp, without producing a consistency failure. This can also help when debugging a problem during production, as you know that the version running on the test floor is exactly the version you released.

Consistency protection does not prevent unauthorized changes, but it prevents use of a modified test program. IG-XL refuses to run a protected test program if the test program has been modified. Consistency protection can detect the following:

- Cell content changes in a worksheet

- Any changes in a pattern

- Edits to any VB modules

Test program protection requires that the test program run online and in Production mode. You cannot run a protected program offline or in Engineering mode.

☑ Note:

Protecting a test program is a one-way process. Once you protect a test program, you cannot “unprotect” it or retrieve the original, modifiable version of the test program. For this reason, always maintain the original, unprotected version of every test program you protect.

Two pieces of software participate in protecting a test program:

- The Test Program Protection Tool generates a protected test program from a set of source files.

- IG-XL then interprets the protected test program at load time and validates and runs the program normally.

The following IG-XL tools honor test program protection by returning an error and refusing to operate on protected files:

- PatternTool

- Pattern Reverse Compiler

- DataTool’s Import New Workbook and Import Worksheet functions

After all of a test program’s files and patterns have loaded, protected test programs show no difference in test time relative to unprotected test programs. Similarly, accuracy and output data are unaffected.

Once you load a protected test program, you can validate and run it as usual.

When an upgrade of IG-XL applies required updates to a test program the first time you open the test program in the new version of IG-XL, you do not need to take any action. IG-XL handles these updates without requiring changes to your protected test program. However, if upgrading IG-XL requires that you make manual changes to your test program, such as changing VBT code or recompiling a digital pattern, you must apply these changes to the unprotected test program, revalidate the program, and reprotect the updated program.

The Test Program Protection Tool does not require a license.

This document covers the following topics:

- Protecting a Test Program
- Running a Protected Test Program
- Working With Test Program Files